

Essays

Stage 1

Pass Exam 1 first. Understand the instructions in the course handbook. Develop two problem descriptions from the [handbook list](#) or from your direct communication with the teacher. Classify the problems according to the ontology so that they are in different categories ([quick reminder](#), for definitions see Maps&Lists). Submit each problem to an AI and check that the response is relevant. Then submit here the same descriptions, and include at the end of each the original number from the topic list and the class number in this way: *topic #99, class 9*. (In case of teacher-approved topic, mark *topic #0*). The teacher will either reject (with comments) and let you start over, or approve and write feedback in this style: "Do *some sort* of specification to the AI request in order to obtain *certain kind* of response."

Ontology with three categories and 11 classes [Close ontology](#)

- "Insecurity": What can happen if there is a lack of security, what can then be done, and how can everything be understood (abstractions)?
 1. Information and threats
 2. Abstract security
 3. Lost security
- How are people situated in the field of security?
 4. Society
 5. Individuals & groups
 6. Organizations
- How do technologies help?
 7. Secure practices
 8. Crypto
 9. Tangible security
 10. Transporting data
 11. Processing and storing

[Close ontology](#)

The same list of essay topics as in the handbook [Close list](#)

- #1. Components of current-day spreading malware.
- #2. Current state of macro viruses.
- #3. How can malware be detected if it has never been seen before?

- #4. Reverse engineering of malware code.
- #5. Evasion techniques used by malware.
- #6. How does the digital immune system work among a large user base of antivirus software?
- #7. Money ecosystem of cyber criminals.
- #8. How can an intruder get into an information system without credentials?
- #9. How can an intruder stay “safe” inside an information system?
- #10. Psychological explanation of “hacker’s ethics”.
- #11. Opportunities for an identity thief.
- #12. How has social engineering developed over time (since, say, year 2000)?
- #13. Effect of training and awareness against social engineering.
- #14. Recent development in user experience (a.k.a., usability) in cyber security.
- #15. A summary of legal code in a chosen area of cyber security in your country.
- #16. Forensics and the possibilities of attribution in case of malicious information operations.
- #17. A chosen type of failures to comply with GDPR, explained with examples of penalties.
- #18. Sanitization in sociological research.
- #19. Does AI help more the internet trolls or those who fight against misinformation?
- #20. How can settings on mobile apps be automated to match the user’s security preferences?
- #21. Preventing data leakage from an organization by filters.
- #22. The costs and benefits of an information security management system (ISMS).
- #23. Guidelines or standards for secure information processing practices in your home country.
- #24. End-to-end encryption for mobile phone calls.
- #25. Identity-based encryption, idea, practicality, and applications.
- #26. Homomorphic encryption, idea, practicality, and applications.
- #27. How is the NTRU cryptosystem doing in the market? What are its benefits and drawbacks?
- #28. Contract signing – when the parties are remote.
- #29. How does technology help in digital rights management (DRM)?
- #30. Parameters for TLS setup (i.e. not for a handshake but for an installation).
- #31. What can go wrong at different parts of the life cycle of symmetric cryptographic keys?
- #32. Various levels of PKI certification policies.
- #33. Blockchain in other uses than cryptocurrency.
- #34. Properties of parental control software.
- #35. Technical properties of password managers.
- #36. Properties of a chosen class of security software for end-users.
- #37. Breaking anonymity in TOR.
- #38. Approaches to censoring the citizens' internet usage.
- #39. Ability of authorities to access mobile phone traffic data.
- #40. Tracking internet users.
- #41. A variety of unlock mechanisms for mobile phones.
- #42. Body implants / prosthesis that communicate.

- #43. Low-level IoT gadgets.
- #44. Anti-tampering technologies on users' devices.
- #45. Examples of successful product certifications against Common Criteria or a similar standard.
- #46. Long-term storage of digital archives.
- #47. Protecting against disturbance or data leakage by electromagnetic radiation.
- #48. Web browser as a substitute of an operating system.
- #49. Hardware-assisted protections in modern operating systems.
- #50. APP-ification in the sense of web transactions being moved to mobile apps.
- #51. API-fication in the sense of non-professional software developers.
- #52. Security techniques at various layers of a database management system.
- #53. What properties in software development environments help in preventing insecure code?
- #54. How is software tested against security vulnerabilities?
- #55. Gamification of learning environments (either: cheating such a system, or: learning security topics).
- #56. Virtual economies inside games or apps, new vistas for criminals?
- #57. Misinformation as a tool for far-reaching purposes.

These are the Exam 3 questions, which need clear modification to serve as essay topics. You need not but in some you may even change the aspect. Now the current classification is given in the parentheses.

- #101. Describe the threat landscape of using an unprotected wireless network. Explain first what it means that a wireless network is unprotected. (1)
- #102. Protecting an information system (IS) against threats directly requires some security measures. Besides such requirements a cyber security professional (CybSecP) must consider requirements that come from outside the system itself. What kinds of requirements can there be? (2)
- #103. What kinds of jobs are there for information security professionals? In other words, how are security activities divided among various occupations, that have security as their main task? (2)
- #104. How can cyber security be tested, and what are the benefits and challenges of doing so? (2)
- #105. Cities are getting smarter, and the enabling technology is IoT, Internet of Things. How should different stakeholders of smart cities deal with cyber security? (2)
- #106. Ransomware: what makes it possible, how does it operate, how to prevent it, how to recover from it? (3)
- #107. What are the seven critical sectors of a society that cyber security should protect? Give real or invented examples of cyber attacks, concentrating on the attack vectors and not the consequences. (4)
- #108. What makes elderly people risky with respect to cyber security? What ways are there to improve the situation? (5)
- #109. How and why should an organization's password policy take into account human factors? (5)
- #110. First describe motivations of cyber criminals. Then describe the opportunities and difficulties they have in reaching their goals. (5)
- #111. What are special requirements that should be expressed in organizational security policies with respect to remote work in contrast to on-site work? You can, but you need not, take into account that these requirements may have been put up rather quickly because of the pandemic situation (in 2020). (6)
- #112. Humans can be authenticated more strongly to information systems than with passwords, but there are weaknesses also in strong methods. What are the stronger methods and their weaknesses? (7)
- #113. Describe requirements, opportunities and challenges of online voting. (8)
- #114. What are the physical threats to information security? (9)

#115. How can voice recognition be used for security purposes, and what can go wrong? (9)

#116. Describe secure networking solutions to extensive remote work. (10)

Close list

Stage 1 Topic 1

Nothing submitted yet.

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Stage 1 Topic 2

Nothing submitted yet.

There is space for characters.

☐ Ready to submit?

SAVE (Check "Ready" if you are done.)

Check "Ready" only when you are done with **both** topics! You cannot get back to this stage afterwards.